Red Hat OpenShift Administration II: Configuring a Production Cluster

Course | Lab Environment

Lab: Manage Declarative Resources

Deploy and update applications from resource manifests that are parameterized for different target environments.

Outcomes

- Deploy an application by using Kustomize from provided files.
- Apply an application update that changes a deployment.
- Deploy an overlay of the application that increases the number of replicas.

As the `student` user on the `workstation` machine, use the `lab` command to prepare your environment for this exercise, and to ensure that all required resources are available:

```
[student@workstation ~]$ lab start declarative-review
```

Instructions

The API URL of your OpenShift cluster is `https://api.lab.example.com:6443`, and the `oc` command is already installed on your `workstation` machine.

Log in to the OpenShift cluster as the `developer` user with `developer` as the password.

1. Clone the v1.1.0 version of the application from the `https://git.lab.example.com/developer/declarative-review.git` URL. Because this repository uses Git branches to represent application versions, you must use the v1.1.0 branch.

   1.1. Clone the v1.1.0 version of the application.

   ```
   [student@workstation ~]$ git clone \
   https://git.lab.example.com/developer/declarative-review.git --branch
   v1.1.0
   Cloning into 'declarative-review'...
   ...output omitted...
   ```

   1.2. Change to the repository directory.

   ```
   [student@workstation ~]$ cd declarative-review
   ```

   Hide Solution

2. Examine the v1.1.0 version of the application. The application consists of a web application that relies on a database.

   2.1. Use the `tree` command to list the structure of the repository.

   ```
   [student@workstation declarative-review]$ tree
   .
   ├── README.md
   ├── base
   │   ├── database  (1)
   │   │   ├── configmap.yaml
   │   │   ├── deployment.yaml
   │   │   ├── kustomization.yaml
   │   │   └── service.yaml
   │   ├── exoplanets  (2)
   │   │   ├── deployment.yaml
   │   │   ├── kustomization.yaml
   │   │   ├── route.yaml
   │   │   └── service.yaml
   │   └── kustomization.yaml  (3)
   └── overlays  (4)
       └── production
           ├── kustomization.yaml
           └── patch-replicas.yaml

   6 directories, 12 files
   ```

1. The `database` base defines resources to deploy a database.
2. The `exoplanets` base defines resources to deploy an application that uses the database.
3. The repository has a `kustomization.yaml` file at the root, which uses two other bases.
4. The repository also has a `production` overlay.

2.2. Examine the `base/kustomization.yaml` file.

```
---
kind: Kustomization
resources:
  - database
  - exoplanets
secretGenerator:
  - name: db-secrets
    literals:
      - DB_ADMIN_PASSWORD=postgres
      - DB_NAME=database
      - DB_PASSWORD=password
      - DB_USER=user
configMapGenerator:
  - name: db-config
    literals:
      - DB_HOST=database
      - DB_PORT=5432
```

The `base/kustomization.yaml` file uses the `database` and `exoplanets` bases, and declares the database connection parameters in a secret resource and a configuration map.

Hide Solution

3. Log in to the OpenShift cluster as the `developer` user with `developer` as the password. Deploy the `base` directory of the repository to a new `declarative-review` project. Verify that the v1.1.0 version of the application is available at `https://exoplanets-declarative-review.apps.lab.example.com`

3.1. Log in to the OpenShift cluster as the `developer` user with `developer` as the password.

```
[student@workstation declarative-review]$ oc login -u developer -p de
veloper \
https://api.lab.example.com:6443
Login successful.

...output omitted...
```

3.2. Create the `declarative-review` project.

```
[student@workstation declarative-review]$ oc new-project declarative-
review
...output omitted...
```

3.3. Use the `oc apply` command to deploy the application. Use the `-k` option to use the Kustomize tool.

```
[student@workstation declarative-review]$ oc apply -k base
configmap/database created
configmap/db-config-2d7thbcgkc created
secret/db-secrets-55cbgc8c6m created
service/database created
service/exoplanets created
deployment.apps/database created
deployment.apps/exoplanets created
route.route.openshift.io/exoplanets created
```

3.4. Use the `watch` command to wait until the workloads are running. The `exoplanets` pod might fail at first, because the database is not ready and the connection fails. You can safely ignore the warning message.

```
[student@workstation declarative-review]$ watch oc get all
Every 2.0s: oc get all                                  ...

Warning: apps.openshift.io/v1 DeploymentConfig is deprecated in v4.14
+, unavailable no sooner than v6.0+
NAME                              READY   STATUS    RESTARTS       AG
E
pod/database-5f64bb7c5-dh7lk      1/1     Running   0              2m
23s
pod/exoplanets-7c48c495d4-ds9r9   1/1     Running   3 (2m1s ago)   2m
23s

NAME                 TYPE        CLUSTER-IP       EXTERNAL-IP   PORT
(S)    AGE
service/database     ClusterIP   172.30.116.226   <none>        5432/
TCP   2m23s
service/exoplanets   ClusterIP   172.30.238.192   <none>        8080/
TCP   2m23s

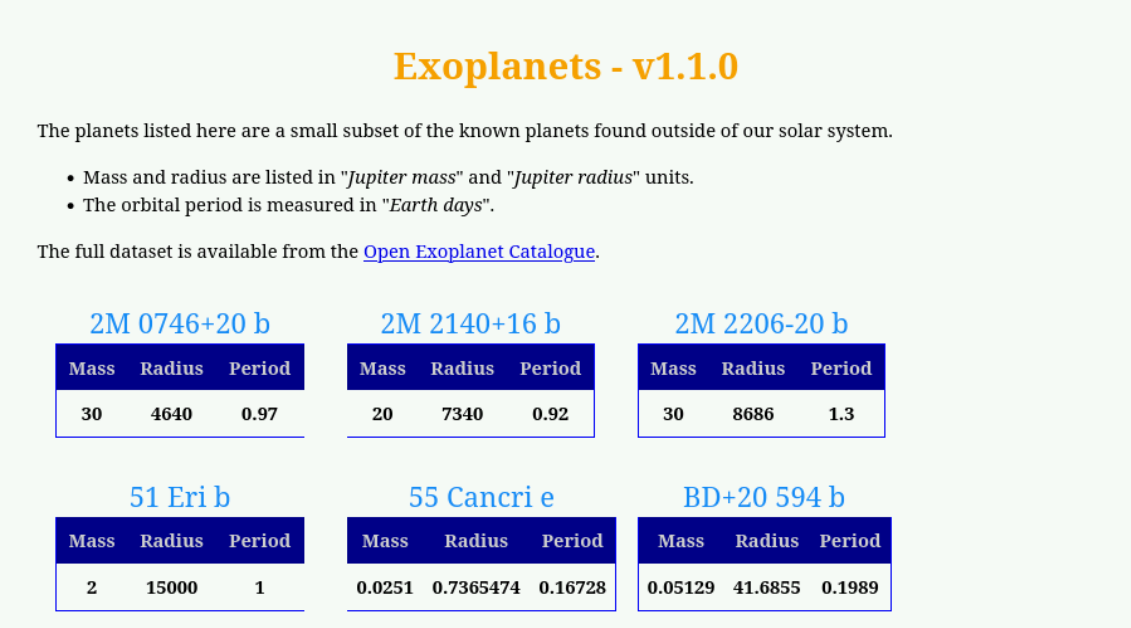
NAME                         READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/database     1/1     1            1           2m23s
deployment.apps/exoplanets   1/1     1            1           2m23s

NAME                                    DESIRED   CURRENT   READY   A
GE
replicaset.apps/database-5f64bb7c5      1         1         1       2
m23s
replicaset.apps/exoplanets-7c48c495d4   1         1         1       2
m23s

NAME                                  HOST/PORT
...
route.route.openshift.io/exoplanets   exoplanets-declarative-review.a
pps.lab.example.com   ...
```

Press **Ctrl**+**C** to exit the `watch` command.

3.5. Open a web browser and go to `https://exoplanets-declarative-review.apps.lab.example.com`



Exoplanets - v1.1.0

The planets listed here are a small subset of the known planets found outside of our solar system.

- Mass and radius are listed in "*Jupiter mass*" and "*Jupiter radius*" units.
- The orbital period is measured in "*Earth days*".

The full dataset is available from the Open Exoplanet Catalogue.

2M 0746+20 b

Mass	Radius	Period
30	4640	0.97

2M 2140+16 b

Mass	Radius	Period
20	7340	0.92

2M 2206-20 b

Mass	Radius	Period
30	8686	1.3

51 Eri b

Mass	Radius	Period
2	15000	1

55 Cancri e

Mass	Radius	Period
0.0251	0.7365474	0.16728

BD+20 594 b

Mass	Radius	Period
0.05129	41.6855	0.1989

The browser displays version v1.1.0 of the application.

Hide Solution

4. Change to the v1.1.1 version of the application by using the v1.1.1 branch of the repository. Examine the changes.

4.1. Change to the v1.1.1 branch.

```
[student@workstation declarative-review]$ git checkout v1.1.1
branch 'v1.1.1' set up to track 'origin/v1.1.1'.
Switched to a new branch 'v1.1.1'
```

4.2. Use the `git show` command to display the last commit.

```
[student@workstation declarative-review]$ git show
...output omitted...
diff --git a/base/exoplanets/deployment.yaml b/base/exoplanets/deploy
ment.yaml
index 66b25cd..1800469 100644
--- a/base/exoplanets/deployment.yaml
+++ b/base/exoplanets/deployment.yaml
@@ -24,7 +24,7 @@ spec:
               name: db-config
           - secretRef:
               name: db-secrets
-        image: registry.lab.example.com:8443/redhattraining/exoplane
ts:v1.1.0
+        image: registry.lab.example.com:8443/redhattraining/exoplane
ts:v1.1.1
         imagePullPolicy: Always
         livenessProbe:
           httpGet:
```

The v1.1.1 version updates the application to the v1.1.1 image.

Hide Solution

5. Deploy the updated application and verify that the URL now displays the v1.1.1 version.

5.1. Use the `oc apply -k` command to apply the changes.

```
[student@workstation declarative-review]$ oc apply -k base
...output omitted...
```

5.2. Use the `watch` command to wait until the application redeploys.

```
[student@workstation declarative-review]$ watch oc get all
Every 2.0s: oc get all                                  ...

Warning: apps.openshift.io/v1 DeploymentConfig is deprecated in v4.14
+, unavailable no sooner than v6.0+
NAME                              READY   STATUS    RESTARTS   AGE
pod/database-5f64bb7c5-dh7lk      1/1     Running   0          14m
pod/exoplanets-5c98f7688f-p4rgm   1/1     Running   0          38s

NAME                 TYPE        CLUSTER-IP       EXTERNAL-IP   PORT
(S)    AGE
service/database     ClusterIP   172.30.116.226   <none>        5432/
TCP   14m
service/exoplanets   ClusterIP   172.30.238.192   <none>        8080/
TCP   14m

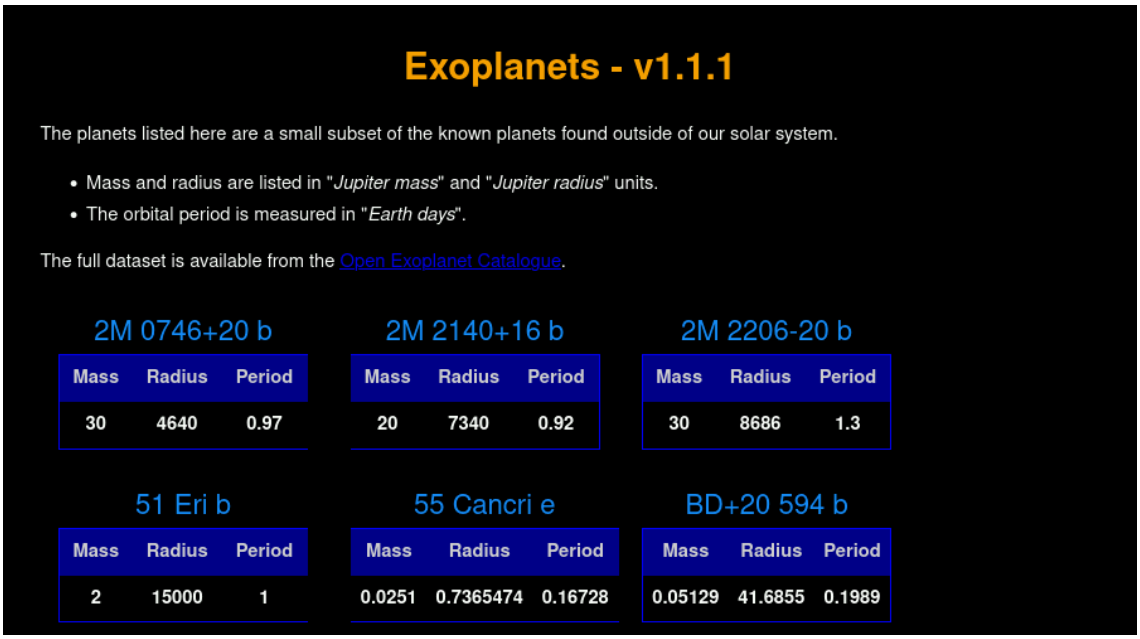
NAME                         READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/database     1/1     1            1           14m
deployment.apps/exoplanets   1/1     1            1           14m

NAME                                    DESIRED   CURRENT   READY   A
GE
replicaset.apps/database-5f64bb7c5      1         1         1       1
4m
replicaset.apps/exoplanets-5c98f7688f   1         1         1       3
8s
replicaset.apps/exoplanets-7c48c495d4   0         0         0       1
4m

NAME                                  HOST/PORT
...
route.route.openshift.io/exoplanets   exoplanets-declarative-review.a
pps.lab.example.com    ...
```

Press **Ctrl+C** to exit the `watch` command.

5.3. Refresh the web browser at `https://exoplanets-declarative-review.apps.lab.example.com`



The browser displays version v1.1.1 of the application.

Hide Solution

6. Examine the overlay in the `overlays/production` directory.

6.1. Examine the `overlays/production/kustomization.yaml` file.

```
---
kind: Kustomization
resources:
  - ../../base/
patches:
  - path: patch-replicas.yaml
    target:
      kind: Deployment
      name: exoplanets
```

This overlay applies a patch over the base.

6.2. Examine the `overlays/production/patch-replicas.yaml` file.

```
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: exoplanets
spec:
  replicas: 2
```

This patch increases the number of replicas of the deployment, so that the production deployment can handle more users.

Hide Solution

7. Deploy the production overlay to a new `declarative-review-production` project. Verify that the v1.1.1 version of the application is available at `https://exoplanets-declarative-review-production.apps.lab.example.com` with two replicas.

7.1. Create the `declarative-review-production` project.

```
[student@workstation declarative-review]$ oc new-project declarative-
review-production
Now using project "declarative-review-production" on server "https://
api.lab.example.com:6443".
...output omitted...
```

7.2. Use the `oc apply -k` command to deploy the overlay.

```
[student@workstation declarative-review]$ oc apply -k overlays/produc
tion
configmap/database created
configmap/db-config-2d7thbcgkc created
secret/db-secrets-55cbgc8c6m created
service/database created
service/exoplanets created
deployment.apps/database created
deployment.apps/exoplanets created
route.route.openshift.io/exoplanets created
```

7.3. Use the `watch` command to wait until the workloads are running.

```
[student@workstation declarative-review]$ watch oc get all
Every 2.0s: oc get all                                  ...

Warning: apps.openshift.io/v1 DeploymentConfig is deprecated in v4.14
+, unavailable no sooner than v6.0+
NAME                              READY   STATUS    RESTARTS      AGE
pod/database-5f64bb7c5-wmgq5      1/1     Running   0             56s
pod/exoplanets-5c98f7688f-kt84d   1/1     Running   3 (41s ago)   56s
pod/exoplanets-5c98f7688f-tlfbt   1/1     Running   2 (53s ago)   56s

NAME                 TYPE        CLUSTER-IP      EXTERNAL-IP   PORT
(S)    AGE
service/database     ClusterIP   172.30.222.83   <none>        5432/T
CP   56s
service/exoplanets   ClusterIP   172.30.46.34    <none>        8080/T
CP   56s

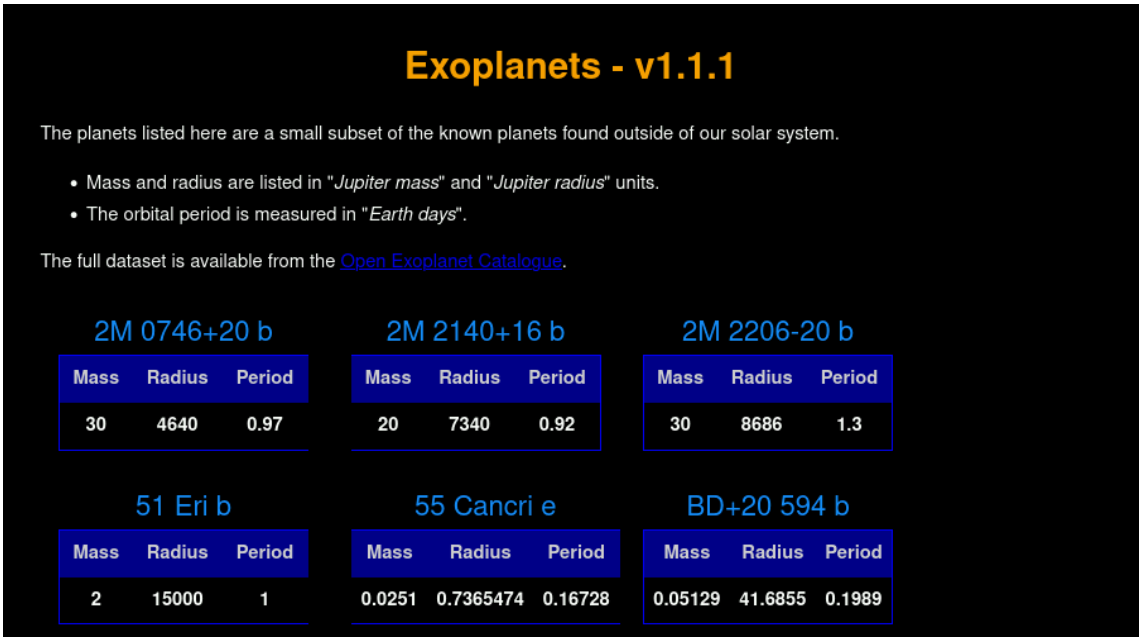
NAME                         READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/database     1/1     1            1           56s
deployment.apps/exoplanets   2/2     2            2           56s

NAME                                    DESIRED   CURRENT   READY   A
GE
replicaset.apps/database-5f64bb7c5      1         1         1       5
6s
replicaset.apps/exoplanets-5c98f7688f   2         2         2       5
6s

NAME                                  HOST/PORT
PATH       ...
route.route.openshift.io/exoplanets   exoplanets-declarative-review-p
roduction.apps.lab.example.com   ...
```

The `exoplanets` deployment has two replicas.

7.4. From the web browser, go to `https://exoplanets-declarative-review-production.apps.lab.example.com`



The browser displays version v1.1.1 of the application.

7.5. Change to the home directory.

```
[student@workstation declarative-review]$ cd
[student@workstation ~]$
```

Hide Solution

Evaluation

As the `student` user on the `workstation` machine, use the `lab` command to grade your work. Correct any reported failures and rerun the command until successful.

```
[student@workstation ~]$ lab grade declarative-review
```

Finish

As the `student` user on the `workstation` machine, use the `lab` command to complete this exercise. This step is important to ensure that resources from previous exercises do not impact upcoming exercises.

```
[student@workstation ~]$ lab finish declarative-review
```

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